

```

3]
class QElement
{
    constructor(element, priority)
    {
        this.element = element;
        this.priority = priority;
    }
}
class PriorityQueue
{
    constructor()
    {
        this.items = [];
    }

    enqueue(element, priority)
    {
        var qElement = new QElement(element, priority);
        var contain = false;
        for (var i = 0; i < this.items.length; i++) {
            if (this.items[i].priority > qElement.priority) {
                this.items.splice(i, 0, qElement);
                contain = true;
                break;
            }
        }

        if (!contain) {
            this.items.push(qElement);
        }
    }

    dequeue()
    {
        if (this.isEmpty())
            return "Underflow";
        return this.items.shift();
    }

    front()
    {
        if (this.isEmpty())
            return "No elements in Queue";
        return this.items[0];
    }

    rear()
    {
        if (this.isEmpty())
            return "No elements in Queue";
        return this.items[this.items.length - 1];
    }

    isEmpty()
    {
        return this.items.length == 0;
    }

    printPQueue()
    {
        var str = "";

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        for (var i = 0; i < this.items.length; i++)
            str += this.items[i].element + " ";
        return str;
    }
}
var priorityQueue = new PriorityQueue();

console.log("Is Queue Empty :",priorityQueue.isEmpty());

console.log(priorityQueue.front());

priorityQueue.enqueue("Pooja", 2);
priorityQueue.enqueue("Aditi", 1);
priorityQueue.enqueue("Atharva", 1);
priorityQueue.enqueue("Madhura", 2);
priorityQueue.enqueue("Devashree", 3);

console.log("Elements of Priority Queue :");
console.log(priorityQueue.printPQueue());

console.log("Front element:",priorityQueue.front().element);

console.log("Rear element :",priorityQueue.rear().element);

console.log("Deleting element :",priorityQueue.dequeue().element);

priorityQueue.enqueue("Sachin", 2);
console.log("New Elements of Priority Queue:");
console.log(priorityQueue.printPQueue());

```

Output -

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Is Queue Empty : true
No elements in Queue
Elements of Priority Queue :
Aditi Atharva Pooja Madhura Devashree
Front element: Aditi
Rear element : Devashree
Deleting element : Aditi
New Elements of Priority Queue:
Atharva Pooja Madhura Sachin Devashree

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